

Predicting Pulsar Stars

For this project, I will be building a classification model that will predict whether a star is a pulsar star or not, based on radio emissions collected. A pulsar is a very highly magnetic, rotating, neutron star, formed from the collapsed core of a star after it's gone supernova, that emits beams of electromagnetic radiation that sweep across space, similar to a lighthouse beacon.

The dataset that will be used is “Predicting Pulsar Star” from Dr Robert Lyon on Kaggle.

This project will begin with data wrangling to prepare the dataset, followed by exploratory data analysis (EDA) to use visualizations to understand the relationships of the predictors. Then after obtaining the training and test sets, a Random Forest classifier will be the model used, and its performance will be evaluated based on F1-score, accuracy, recall, and other metrics, and the project will be concluded with effective communication and visualizations of the findings.